

Bayesian Framework for Post-disruption Travel Time Prediction in Metro Networks

**BIRS workshop on Emerging Challenges in Statistical
Modeling for Transportation Research**

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Motivation

The metro service is resuming following an incident.



How much longer until the next train arrives?

Service interruptions in the metro

STM reports an average of 1K service interruptions lasting more than 5 minutes each year in the Montréal metro.

- Common causes of disruptions include equipment failures, passenger incidents, track obstructions, and operational issues.
- Accurate travel time prediction becomes particularly critical during the recovery phase following a disruption.

Metro and incidents



In 2018, on the Green line:

- around 225 disruptions in each direction,
- lasting on average 11 minutes,
- affecting on average 7.5 trains.

Data source 1 - track occupancy data

- Montréal metro uses a fixed block (sensor) signaling system.
- Rail tracks (platforms and tunnels) are divided into blocks of variable length.
- Sensors record occupation and release of a block.



- The track occupancy data include timestamps (entry/exit), block and train identifiers.
- These are used to reconstruct train journeys (including location and relative distance of all trains).

Data source 2 - disruption logs

A dataset of all the reported incidents in 2018.

STM records location, duration and beginning end of incidents, but

- not all events are reported,
- there are some inaccuracies (resolution times, locations):
we fix these using timestamps of track occupancy data.

Service interruptions

- Disruptions occur unpredictably.
- Even after an incident is officially resolved, service recovery may take a substantial amount of time.
- We aim to predict waiting times during the recovery phase alongside **uncertainty quantification**.

Features of the problem

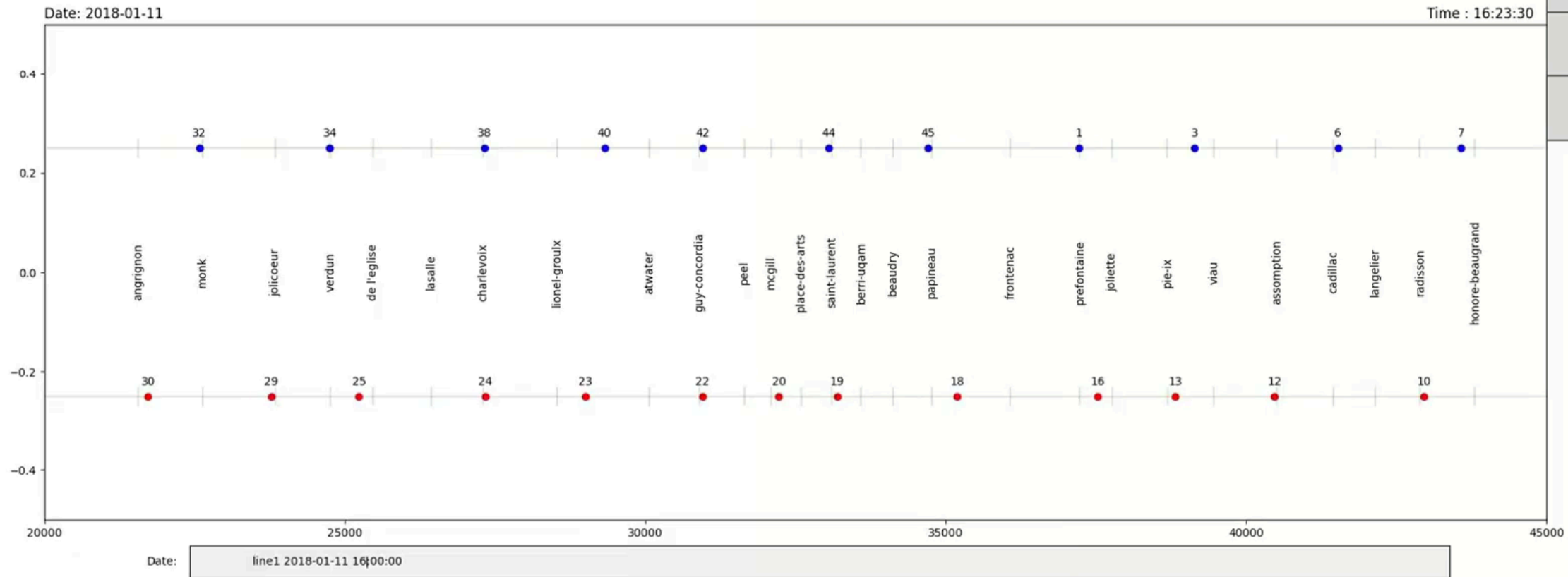
There are relatively few studies on urban metro networks compared to intercity railway systems.

Some (unique) characteristics of the Montréal metro:

- The metro system is a closed-loop (no overtaking).
- No information about passenger flow in real-time.
- Absence of schedule (only average frequency).
- Operational procedures for incidents depending on the time of the day, their duration and underlying cause.

Video of incident

Figure 1

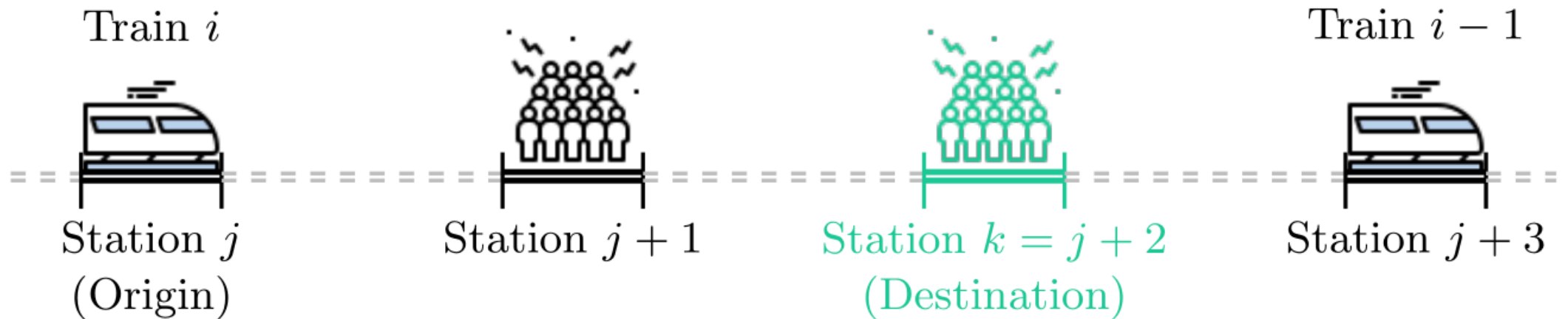


Operational protocols

- Trains can be required to stop immediately at the nearest station.
- Trains can be permitted to continue their journey for several stations before being forced to stop.
- Post-disruption, trains are unevenly spaced, leading to train bunching.
- This leads to passenger accumulation and increased waiting times.

Notation

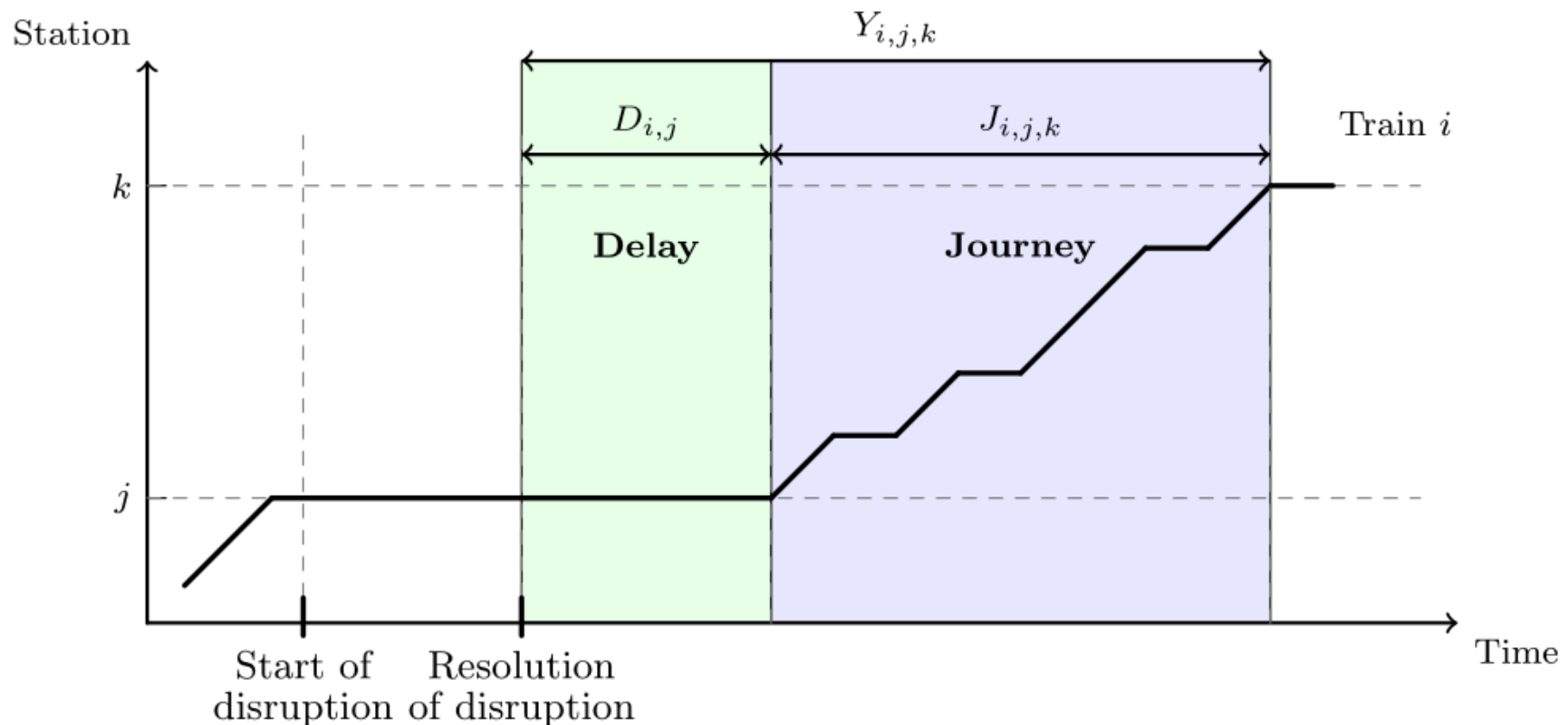
We model the travel time $Y_{i,j,k}(t)$ of train i from origin station j to destination station k at time t .



Decomposition of travel time

$$Y_{i,j,k}(t) = D_{i,j} + J_{i,j,k}(t) + e_{i,j,k}.$$

time
delay
journey
error

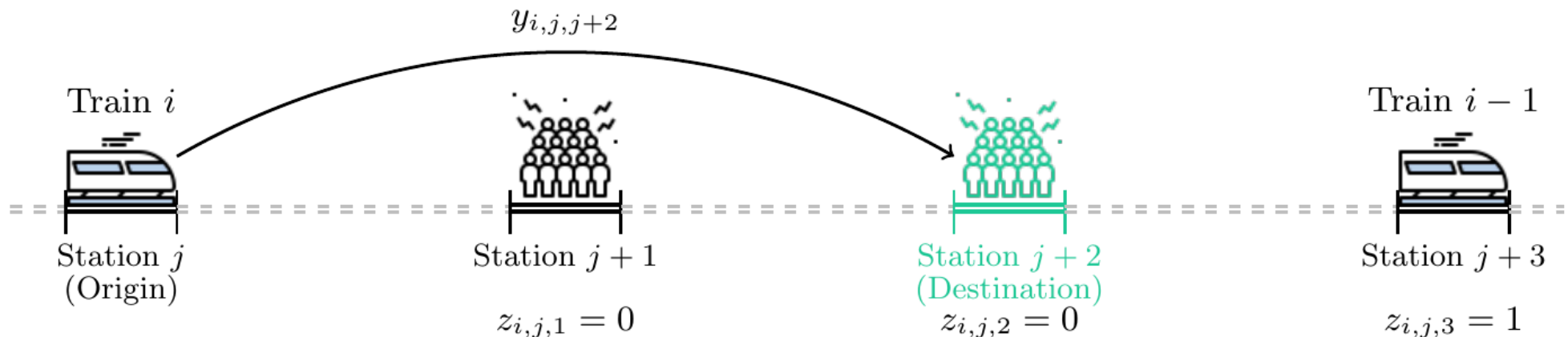


Delay modelling

We suppose that train operations are influenced by the presence of other trains ahead of them in the direction of travel,

$$D_{i,j} = \sum_{l=1}^{D_{\max}} \gamma_{l,j} z_{i,j,l}.$$

Effects $\gamma_{l,j}$ depend on distance ahead (up to $D_{\max} = 5$ stations) and station.



$z_{i,j,l}$: is there is another train l stations ahead when train i was at station j .

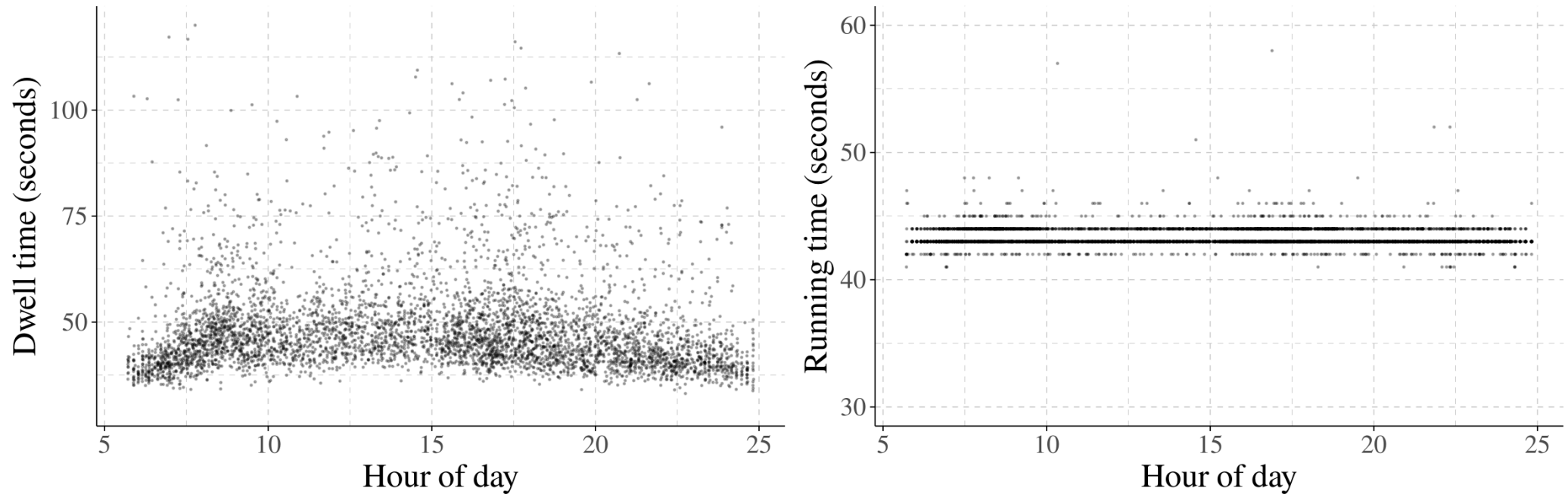
Journey modelling

Journey $J_{i,j,k}(t)$: total time a train spends traversing both platforms and tunnel segments when traveling from origin j to destination k at time t .

Decomposed as the sum of

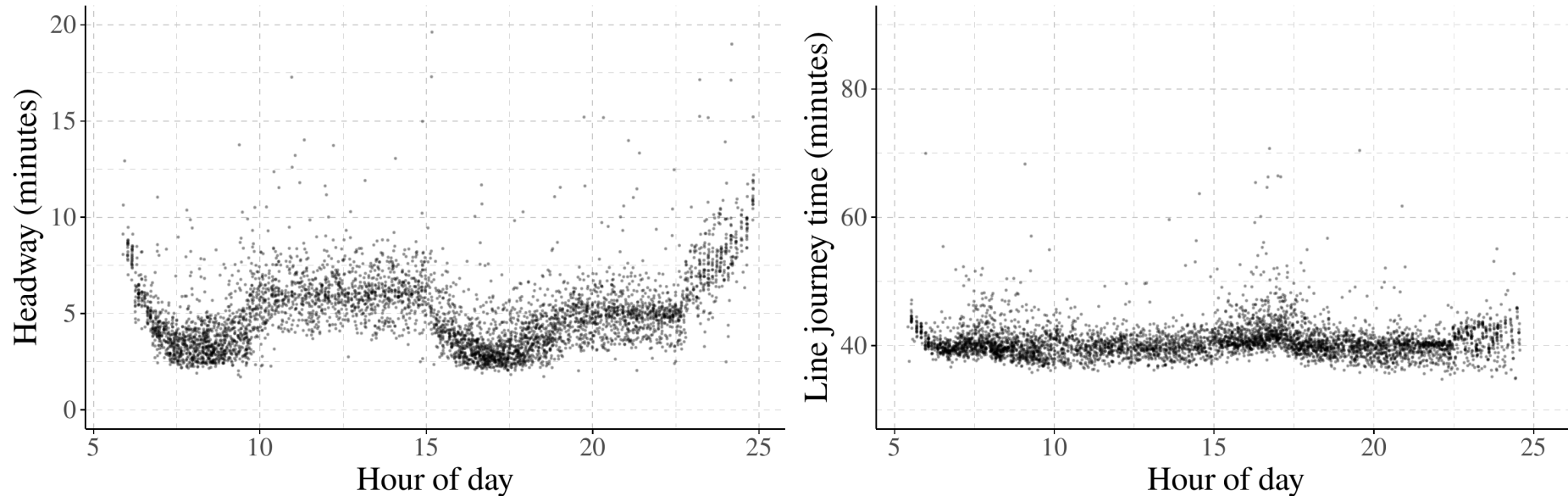
- **dwelt times**: the time a train remains stationary at intermediate stations $j + 1, \dots, k - 1$,
- **running times**: times required to travel between stations (through tunnels).

Dwell times and running times at Guy-Concordia



Running times exhibit relatively low variability because train speeds are tightly regulated by operational constraints.

Headways and travel times under normal operations



Both travel time and headway (temporal gap between two consecutive trains) vary significantly according to the time of the day.

The larger the headway between trains, the more passengers accumulate on the platforms at intermediate stations, leading to increased dwell times.

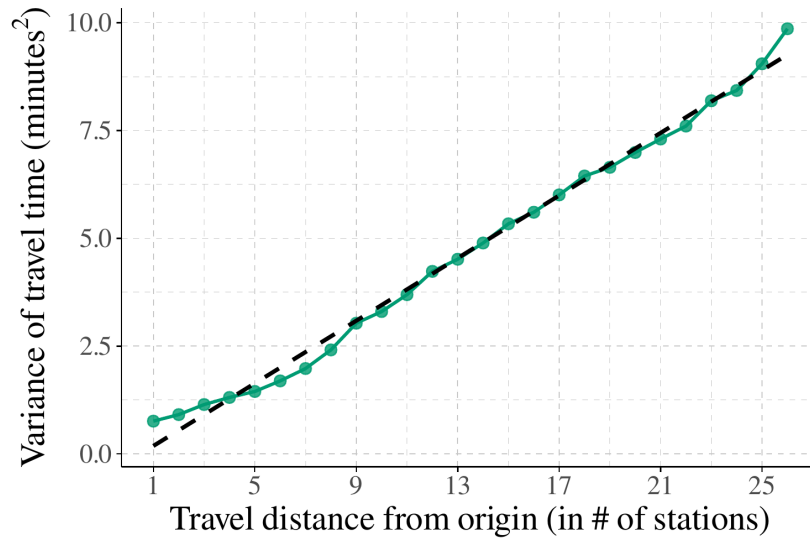
Journey process

We estimate median headway $\tilde{h}_s(t)$ at station s and time t using a moving-window (since passenger flow varies over time of the day).

$$J_{i,j,k}(t) = t_0 + \underbrace{\tilde{Y}_{j,k}(t)}_{\text{median travel time}} + \sum_{s=j+1}^{k-1} \underbrace{\theta_s [h_{i,s} - \tilde{h}_s(t)]}_{\text{difference to median headway}}.$$

For every additional minute of prolonged headway at an intermediary station s , the journey process increases by θ_s .

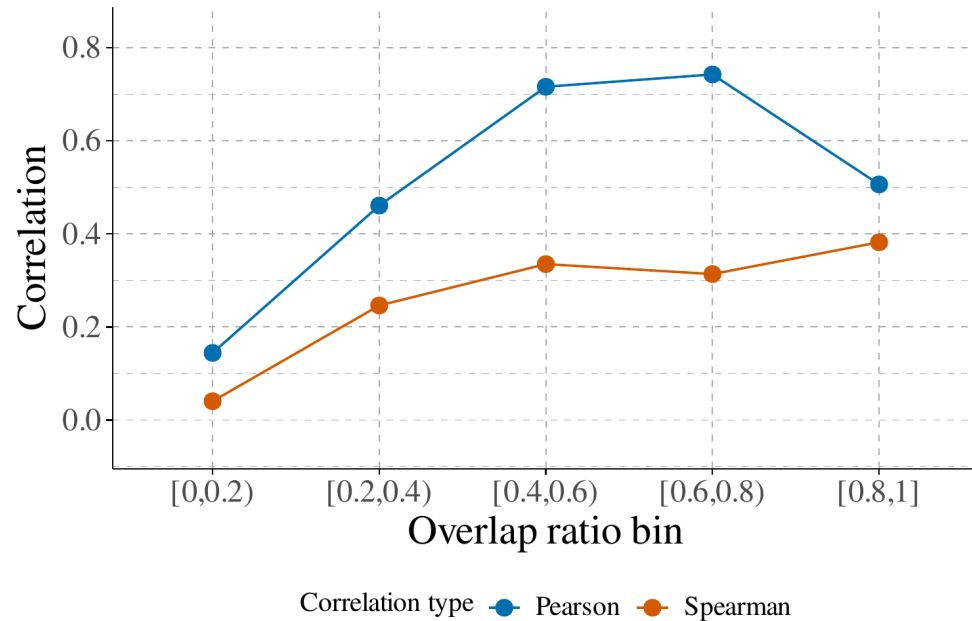
Variance specification



Travel times consist of the sum of running times (mostly identical) and dwell times.

It seems reasonable to hypothesize that the variance increases linearly with the traveled distance, with $\sigma^2 = \omega_0 + \omega_1(k - j)$.

Autocorrelation structure



To account for the dependence between consecutive trains, we use a moving average (MA) that depend on the overlap in journey during post-disruption

$$\Delta_i = (k - j) / (k - j') \in [0, 1),$$

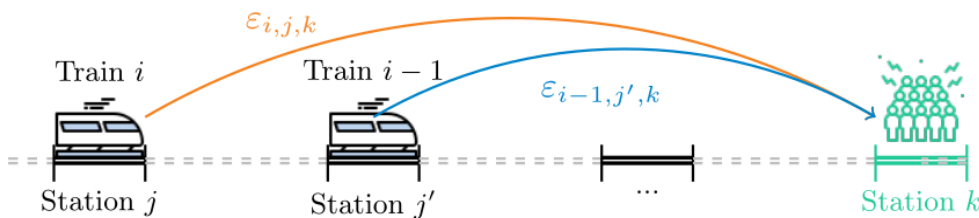
between trains i and $i - 1$ for stations $j < j' < k$.

The error term is written as

$$e_{i,j,k} = \rho_i \varepsilon_{i-1,j',k} + \varepsilon_{i,j,k}$$

with an exponential-type decay,

$$\rho_i = \rho \{1 - \exp(-\lambda \Delta_i)\}.$$



Prediction

The model including delay, journey and error is

$$\underbrace{Y_{i,j,k}(t)}_{\text{travel time}} = \underbrace{D_{i,j}}_{\text{delay}} + \underbrace{J_{i,j,k}(t)}_{\text{journey}} + \underbrace{\rho_i \varepsilon_{i-1,j',k} + \varepsilon_{i,j,k}}_{\text{error}}.$$

The unconditional **predictive distribution** is obtained by integrating out the latent innovation of the preceding train:

$$p(y_{i,j,k}) = \int p(y_{i,j,k} \mid \varepsilon_{i-1,j',k}) p(\varepsilon_{i-1,j',k}) d\varepsilon_{i-1,j',k},$$

which does not admit a closed-form expression.

Error distribution

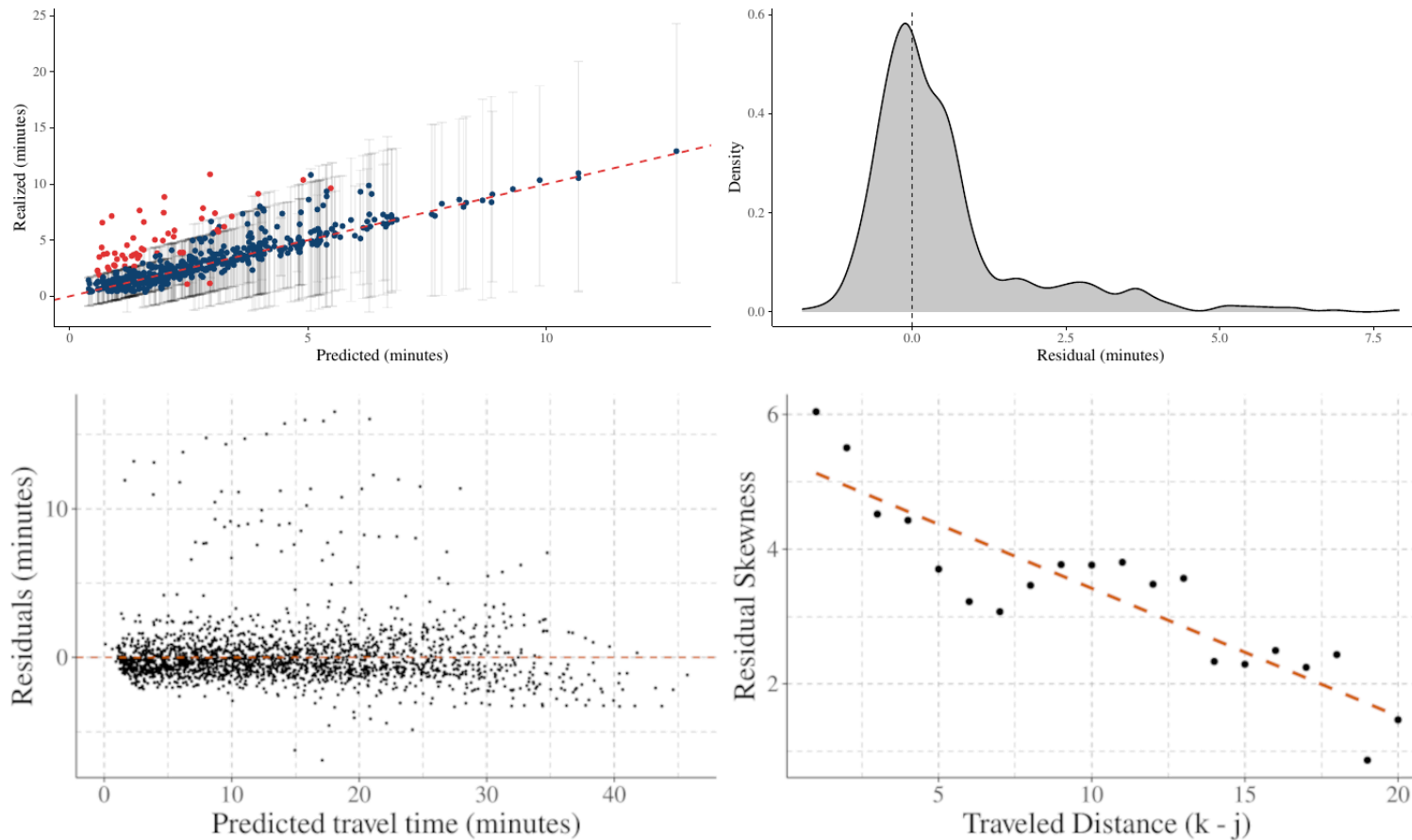
We consider three location-scale distributions for random errors $\varepsilon_{i,j,k}$.

- normal,
- a skew-normal with skewness α ,
- a skew- t with skewness α and degrees of freedom ν .

Both skewed distributions have a mean and variance that are a function of the skewness α (and the degrees of freedom ν for the skew- t).

Distributional assumption

There are several outliers and skewed residuals due to asymmetry: trains can run late, but not really cut on journey times.



We specify the skewness as a linear function $\alpha = \alpha_0 + (k - j)\alpha_1$.

Model estimation and prediction

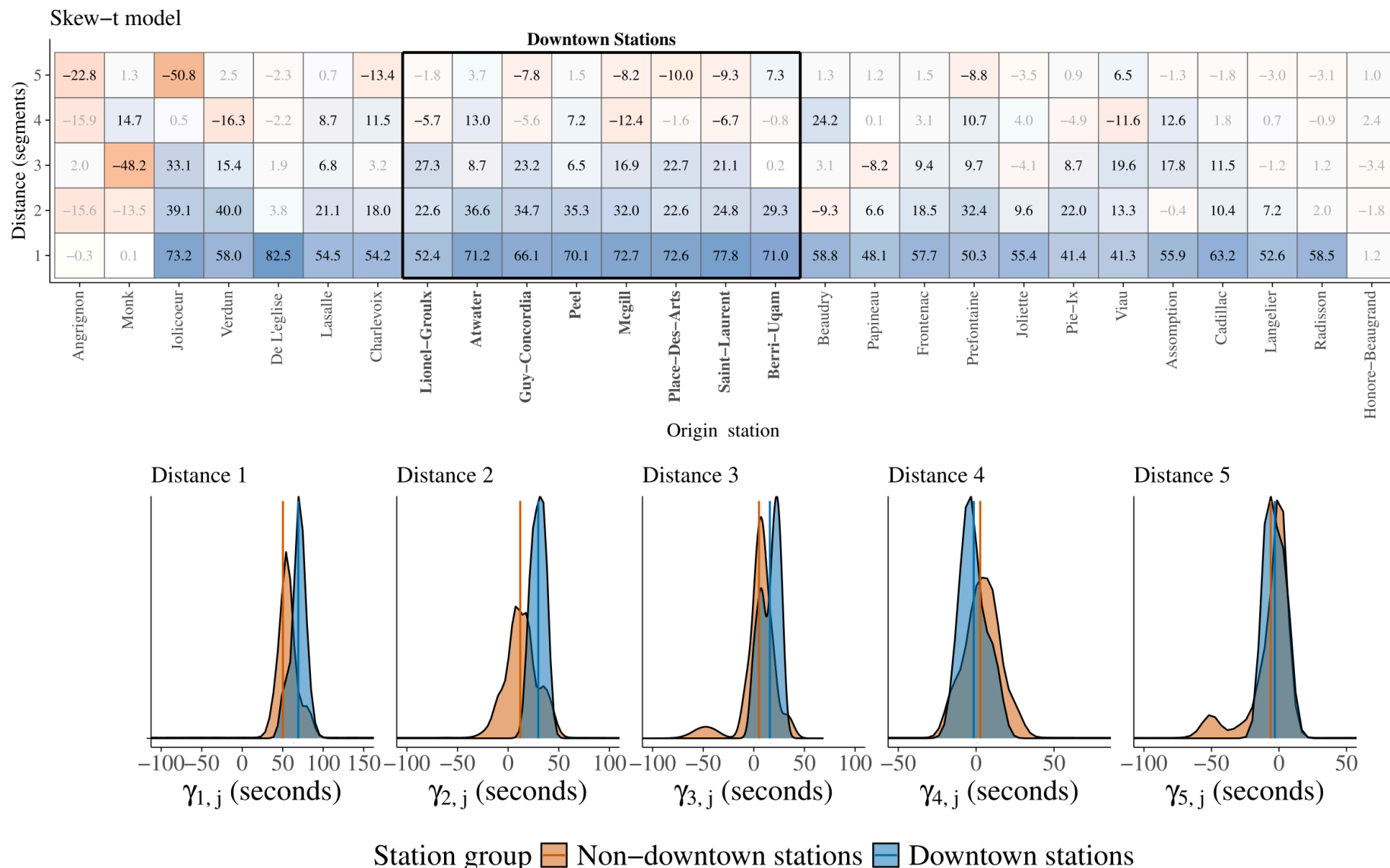
We work in the Bayesian framework and assign weak or vague priors.

We generate posterior draws from the hierarchical model using Hamiltonian Monte Carlo (through **Stan**).

Posterior predictive inference is carried out via simulation by jointly sampling the model parameters and latent errors.

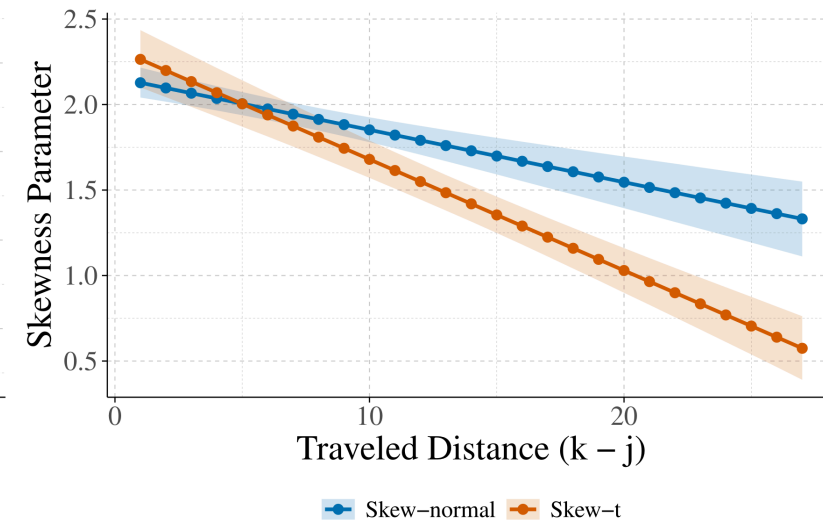
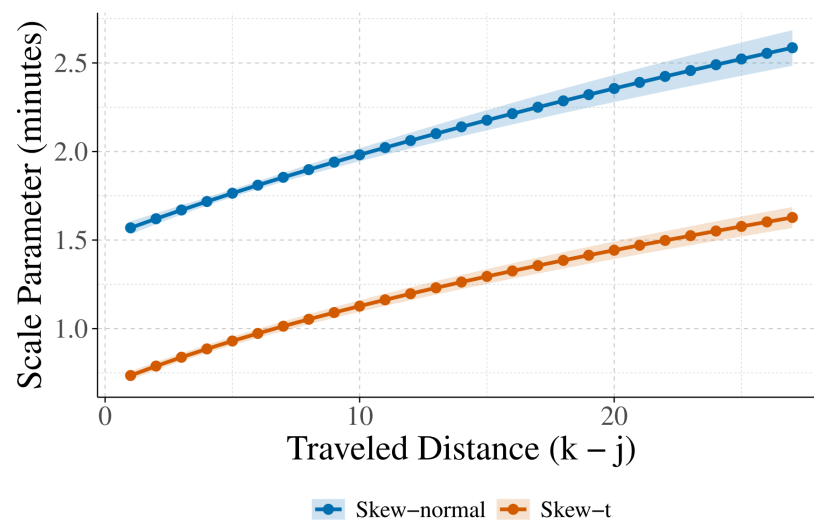
We split the 21K time stamps by disruption into 90% training and 10% hold-out.

Posterior of delay



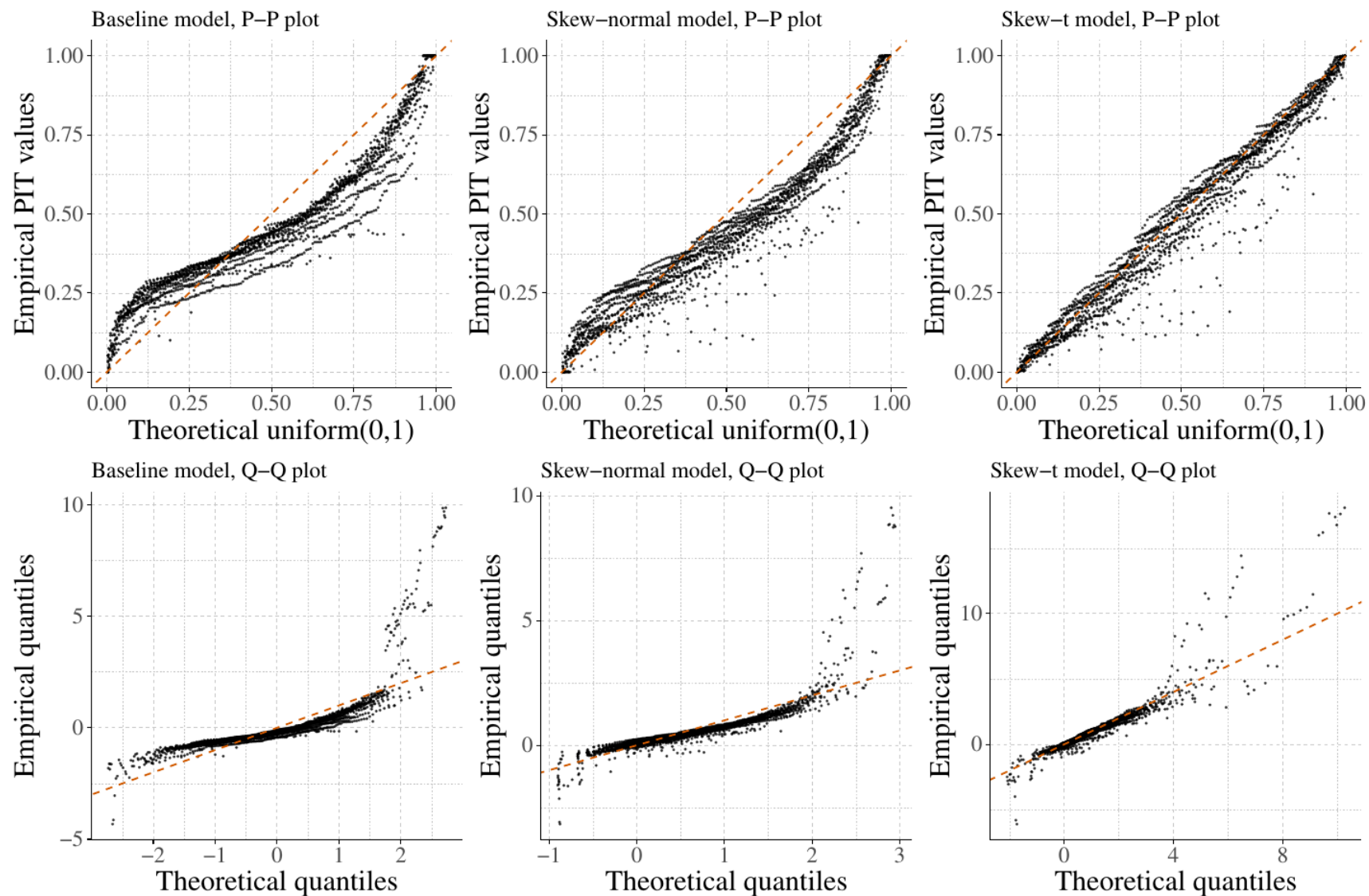
Posterior of parameters

Parameter	Skew-normal			Skew-t		
	Posterior mean	(5%,	95%)	Posterior mean	(5%,	95%)
ω_0	2.300	(	2.168, 2.436)	0.460	(	0.424, 0.497)
ω_1	0.163	(	0.139, 0.185)	0.081	(	0.074, 0.089)
α_0	2.158	(	2.064, 2.254)	2.329	(	2.157, 2.508)
α_1	-0.031	(	-0.041, -0.020)	-0.065	(	-0.076, -0.054)
ν				2.666	(	2.561, 2.773)

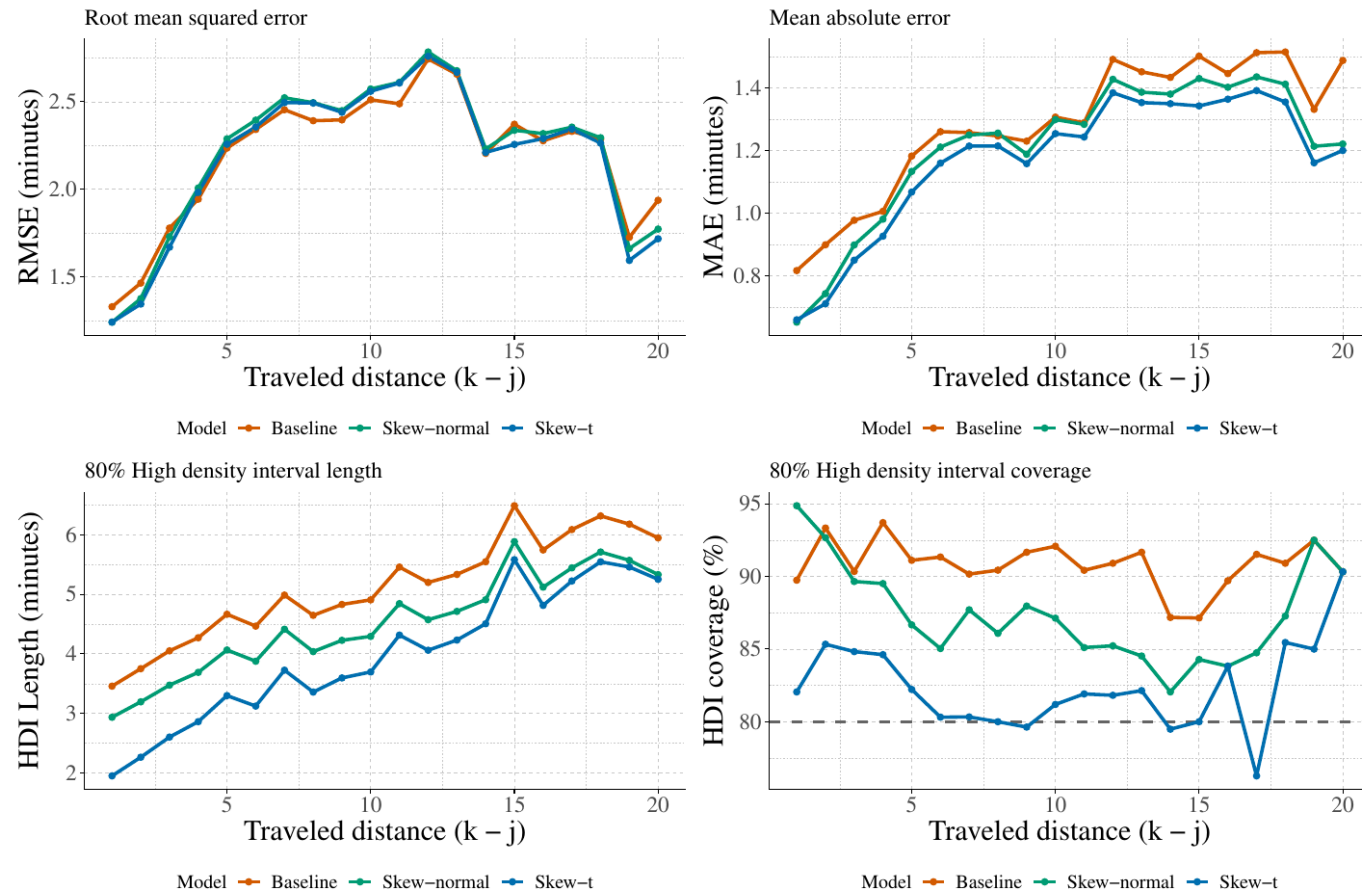


Skewness, scale and degree of freedom parameters.

Goodness-of-fit



Performance metrics



The skewed- t model has better coverage and also lower CRPS values.

Summary

- We examined the problem of predicting post-disruption travel times in an urban metro system.
- We proposed a Bayesian hierarchical model that explicitly accounts for train interactions, headway imbalance due to disruption, and non-Gaussian features in travel time distributions.
- The results provide evidence of meaningful error dependence between consecutive trains.
- Possible extensions include online forecasting setting, in which predictions are continuously updated as new real-time operational information becomes available.

The end

Thank you for your attention. Questions?

Financial support:

Preprint: [arXiv:2602.19952](https://arxiv.org/abs/2602.19952)

